

For example, the Tomboy note-taking application can be launched by running the following from a Python shell:

```
>>> import dbus
>>> bus=dbus.SessionBus()
>>> bus.start_service_by_name('org.gnome.Tomboy')
(True, dbus.UInt32(1L))
```

You can see that Tomboy is started and the function returns True.

In fact, it is very easy to create D-Bus services. Create a text file, called *org.gnome.Newservice.service* for example, with following contents:

```
[D-BUS Service]
Name=org.gnome.Newservice
Exec=/usr/bin/newservice
```

Now you can start Newservice by name.

## Data types and type casting

Since D-Bus is an inter-process message passing mechanism, it deals with various data types, depending on the data to be received or sent. One of the primary benefits of D-Bus is that it is flexible with data type conversions. Since we are more concerned with D-Bus in Python's context, let us take a look at how D-Bus types and Python types are tuned to each other with auto typecasting. D-Bus uses static types. Since Python types and D-Bus types are compatible to each other, we never have to worry about type conversion hurdles.

Table 1 lists the types supported and their conversions. Types marked (\*) may be a subclass of either *int* or *long*, depending on the platform.

From the above table, you can infer that if we have some string to be passed or received through D-Bus daemon, it is received or sent as its equivalent D-Bus type. Likewise, *string* is send as *dbus.String('string')*.

We can call methods provided by the proxy object in two ways—synchronous call or asynchronous call. Synchronous calls block any other methods to be called until the current function call ends and returns something. Asynchronous (non-blocking) method calls allow multiple method calls to be in progress simultaneously, and allow your applications to do other work while it waits for results/answers. Asynchronous calls are invoked by setting up an event loop like *Gmainloop* or *gtk.main()*.

## Hands-on D-Bus client-server

Let us code a simple ExampleObject to be exported under the *org.example.Sample* service and a client application, to understand programming with D-Bus better:

- D-Bus service: *org.example.Sample*
- File name: *dbus-example-service.py*

Lists D-Bus types supported and their conversions

| Python type         | Converted to D-Bus type       | Notes                 |
|---------------------|-------------------------------|-----------------------|
| D-Bus proxy object  | ObjectPath (signature 'o')    | (+)                   |
| dbus.Interface      | ObjectPath (signature 'o')    | (+)                   |
| dbus.service.Object | ObjectPath (signature 'o')    | (+)                   |
| dbus.Boolean        | Boolean (signature 'b')       | a subclass of int     |
| dbus.Byte           | byte (signature 'y')          | a subclass of int     |
| dbus.Int16          | 16-bit signed integer ('n')   | a subclass of int     |
| dbus.Int32          | 32-bit signed integer ('i')   | a subclass of int     |
| dbus.Int64          | 64-bit signed integer ('x')   | (*)                   |
| dbus.UInt16         | 16-bit unsigned integer ('q') | a subclass of int     |
| dbus.UInt32         | 32-bit unsigned integer ('u') | (*)_                  |
| dbus.UInt64         | 64-bit unsigned integer ('t') | (*)_                  |
| dbus.Double         | double-precision float ('d')  | a subclass of float   |
| dbus.ObjectPath     | object path ('o')             | a subclass of str     |
| dbus.Signature      | signature ('g')               | a subclass of str     |
| dbus.String         | string ('s')                  | a subclass of unicode |
| dbus.UTF8String     | string ('s')                  | a subclass of str     |
| bool                | Boolean ('b')                 |                       |
| int or subclass     | 32-bit signed integer ('i')   |                       |
| long or subclass    | 64-bit signed integer ('x')   |                       |
| float or subclass   | double-precision float ('d')  |                       |
| str or subclass     | string ('s')                  | must be valid UTF-8   |
| unicode or subclass | string ('s')                  |                       |

Table 1

```
#!/usr/bin/env python
import gobject
import dbus
import dbus.service
import dbus.mainloop.glib
```

```
class ExampleObject(dbus.service.Object):
    @dbus.service.method('org.example.Sample',
                        in_signature='s', out_signature='as')
    def HelloWorld(self, test_message):
        print (str(test_message))

        return ['Hello World ','dbus-service',str(test_message)]
```

```
@dbus.service.method('org.example.Sample',
                    in_signature='', out_signature='s')
def Ping(self):
    print "Pinged"
    return str("Hi, I am Alive")
```

```
@dbus.service.method('org.example.Sample',
```